

Benchmark validation of the SAF skyrmion micromagnetic simulator

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Abstract

This document records how the synthetic-antiferromagnet (SAF) skyrmion micromagnetic simulator (`src/skyrmion_simulator/simulator`, `src/skyrmion_simulator/phase_diagram`, `src/skyrmion_simulator/stochastic_llgs`) is benchmarked against canonical references, and its current status. Each benchmark wires up the published material parameters and geometry but exercises the *production* physics (field assembly, integrators, torques, demagnetisation, thermal noise). For every case we give the reference, a table comparing the reference material parameters with those actually used, the result figure (this work) next to the reference figure where one exists, and a concise, honest account of the behaviour, any failure, and the limitations. All result figures were regenerated for this document with a square plotting area and no grid; the re-run benchmarks cache their measured data under `output/benchmarks/` so they are computed only once.

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1 Summary

Table 1: Benchmark status overview. Tolerances and deviations are detailed in each section.

#	Benchmark	Reference value	This work	Status
1	Cortés-Ortuño 2018 DMI SP (disc)	$r_{sk} = 22.03$ nm (ODE)	22.61 nm (2.6%)	PASS
2	Bogdanov–Hubert / RT 2013 Eq. 18	$R_s = 9.46$ nm	9.32 nm (1.5%)	PASS
3	NIST muMAG SP#4 (Permalloy)	$\langle m_x \rangle = 0$ at ≈ 136 ps	136.7 ps (0.5%)	PASS
4	NIST muMAG SP#5 (Zhang–Li STT)	$\Delta = (-1.2, -14.7)$ nm	$(-1.21, -12.18)$ nm	PASS
5	Güngördü 2016 phase diagram	FM/SkX/SP/SC regions	FM+SP ok; SkX under-resolved	PASS (qual.)
6	Banerjee 2014 FM $\leftrightarrow$ SkX field	$h_c \approx 1.03$ – 1.52	0.18–0.96 (low)	FAIL
7	RT 2013 Fig. 4(d) confined R_s	≈ 25 nm	27.6 nm (10%)	PASS
8	Rohart 2016 Néel-Arrhenius collapse	Arrhenius; $\Delta E = 26 \pm 4$ meV	$R^2 = 0.99$; $\Delta E = 13.0$ meV	PASS
8b	Rohart 2016 Fig. 5(b) $\Delta E(D)$	rises with D	6.1 $\rightarrow$ 34.4 meV	PASS
9a	Tomasello 2018 size vs T (determ.)	$H=0$ ratio ≈ 3.3	2.52 (converged)	PASS
9b	Tomasello 2018 size vs T (thermal)	ratio ≈ 3.3	4.66 (over-expansion)	FAIL
10	Pham 2024 Fig. S49 Thiele v_{SOT}, v_{TSH}	closed forms	analytic 2×10^{-16} ; LLGS 9.9%	PASS
11	1D domain-wall width	$\Delta = \sqrt{A/K_{\text{eff}}}$	12.637 nm (0.10%)	PASS
12	Uniform-mode FMR frequency	$f = \gamma H_K / 2\pi$	rel. err. $< 5\%$	PASS
14	Free-BC demag kernel	thin slab $N_{zz} \approx 1$	$H_z / \mu_0 M_s = -0.984$	PASS

Guiding rule. The benchmark sets its own parameters and geometry, but the physics it exercises is the production code. Where a continuum micromagnetic run is compared to an *atomistic* reference (#8 vs Rohart 2016) or to a *variational-cell* ansatz (#5/#6 vs Güngördü/Banerjee), the form or trend is gated; grid-dependent absolute numbers are reported but not gated, with the mismatch stated explicitly. Supporting stochastic-infrastructure gates (Néel–Brown reversal, Langevin function, spin-wave equipartition, $T=0$ deterministic-limit) underpin #8 and all pass (§15).

2 Cortés-Ortuño 2018 interfacial-DMI standard problem

Reference. Cortés-Ortuño *et al.*, 2D interfacial (C_{nv}) DMI standard problem on a disc. ODE theory $r_{sk} = 22.03$ nm; OOMMF/Fidimag 21.87; mumax3 22.10. The bundled dataset provides the reference $m_z(r)$ curve.

Table 2: Material parameters (reference vs this work). Identical; no demag (the reference uses NoDemagSpins).

Quantity	Reference (CO 2018)	This work
Exchange A	13 pJ/m	13 pJ/m
DMI D (interfacial)	3 mJ/m ²	3 mJ/m ²
M_s	0.86 MA/m	0.86 MA/m
K_{eff} ($= K_u$)	0.4 MJ/m ³	0.4 MJ/m ³
Disc radius	50 nm	50 nm
Cell / lattice	2 nm, 50^2	2 nm, 50^2
Demagnetisation	off	off

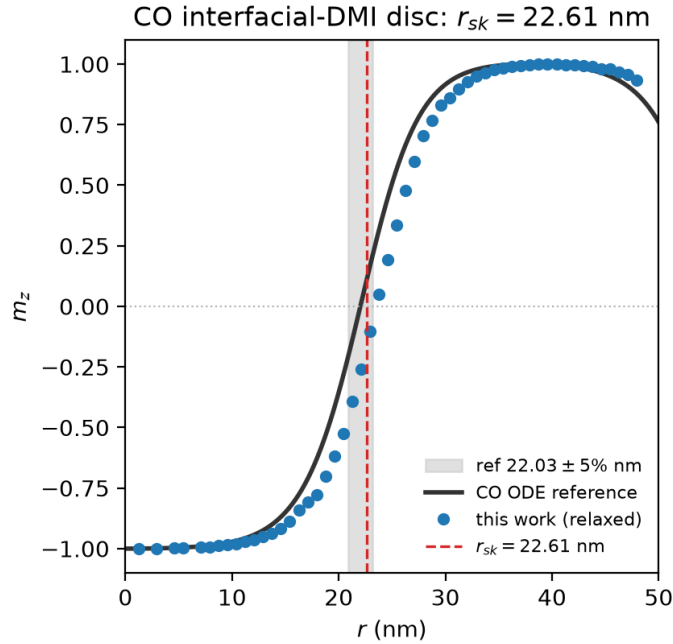


Figure 1: Relaxed radial profile $m_z(r)$ (this work, free-BC disc, production `effective_field+relax`) against the CO ODE reference. Measured $r_{sk} = 22.61$ nm (2.62%, tol 5%); outward-Néel chirality confirmed.

Behaviour / status. **PASS.** $r_{sk} = 22.61$ nm (2.62% vs theory); profile RMS 0.11 (tol 15%).
Limitations: the profile-RMS tolerance is 15% because a 5%-level r_{sk} shift on a $\Delta \approx 5.7$ nm wall already gives RMS ~ 0.1 (wall-position dominated; the published OOMMF/mumax3 runs show the same). The DMI boundary-condition sign follows mumax3 (opposite to RT 2013 Eq. 6) with identical bulk chirality.

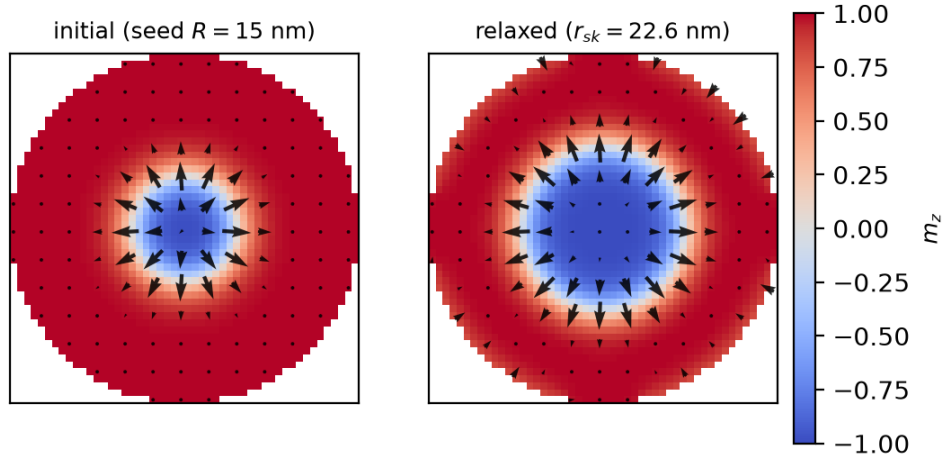


Figure 2: Configurations: the seeded skyrmion ($R = 15$ nm) relaxes to the equilibrium isolated skyrmion ($r_{sk} = 22.6$ nm) in the disc. Colour is m_z ; arrows the in-plane components (outward Néel).

3 Bogdanov–Hubert / RT 2013 isolated-skyrmion profile

Reference. Rohart & Thiaville, *PRB* 88, 184422 (2013), Eq. 18: $R_s = \Delta / \sqrt{2(1 - D/D_c)}$ (infinite film, no demag, K already K_{eff}). $R_{s,\text{ref}} = 9.463$ nm at $D/D_c = 0.825$.

Table 3: Material parameters (reference vs this work).

Quantity	Reference (RT 2013)	This work
Exchange A	16 pJ/m	16 pJ/m
K_{eff}	0.51 MJ/m ³	0.51 MJ/m ³
$\mu_0 M_s$	1.1 T	1.1 T
DMI D	3 mJ/m ²	3 mJ/m ² ($D/D_c=0.825$)
$\Delta = \sqrt{A/K}$	5.60 nm	5.60 nm
Cell / lattice	—	1 nm, 96^2 (PBC)
Demagnetisation	off	off

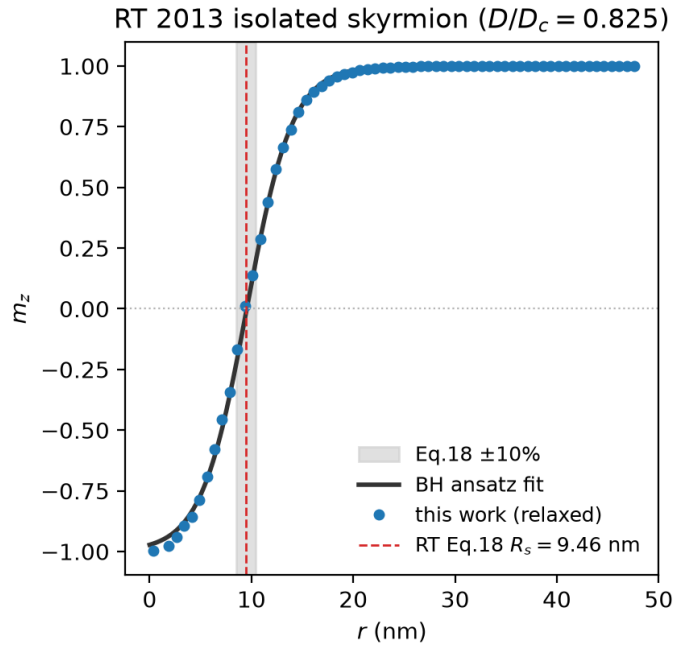


Figure 3: Relaxed $m_z(r)$ (this work) with a Bogdanov–Hubert ansatz fit; the dashed line marks RT Eq. 18 $R_s = 9.46$ nm. Measured $R_s = 9.32$ nm (1.50%, tol 10%), shape RMS 0.012.

Behaviour / status. **PASS.** Only R_s is gated; the fitted wall width $\Delta_{\text{fit}} = 4.53$ nm differs from the thin-wall $\sqrt{A/K}$ because Eq. 18 is itself an asymptotic approximation (curvature-corrected core). Profile RMS reported, not gated.

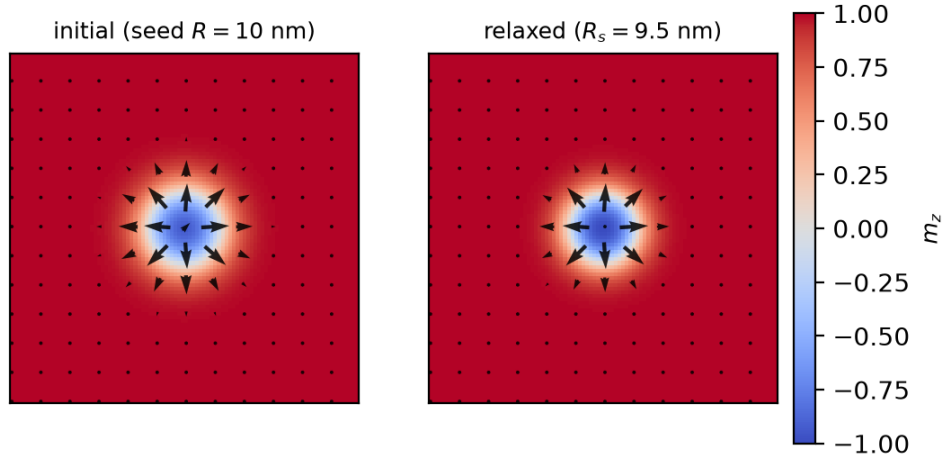


Figure 4: Configurations: seeded skyrmion ($R = 10$ nm) $\rightarrow$ relaxed isolated skyrmion ($R_s = 9.3$ nm) in the PBC box.

4 NIST muMAG Standard Problem #4 (Permalloy, Field 1)

Reference. NIST muMAG SP#4 (Eicke & McMichael). Ten contributed solutions agree to $\sim 3\%$; the first $\langle m_x \rangle = 0$ crossing under Field 1 is at ≈ 136 ps.

Table 4: Material parameters (reference vs this work).

Quantity	Reference (NIST)	This work
Exchange A	13 pJ/m	13 pJ/m
M_s	0.8 MA/m	0.8 MA/m
Anisotropy K	0	0
α	0.02	0.02
γ	1.76×10^{11}	1.76×10^{11}
Geometry	$500 \times 125 \times 3$ nm	same (rect mask)
Field 1 $\mu_0 H$	$(-24.6, 4.3, 0)$ mT	same
Cell	~ 2.5 nm	2.5 nm
Demagnetisation	full	Newell free-BC (single layer)

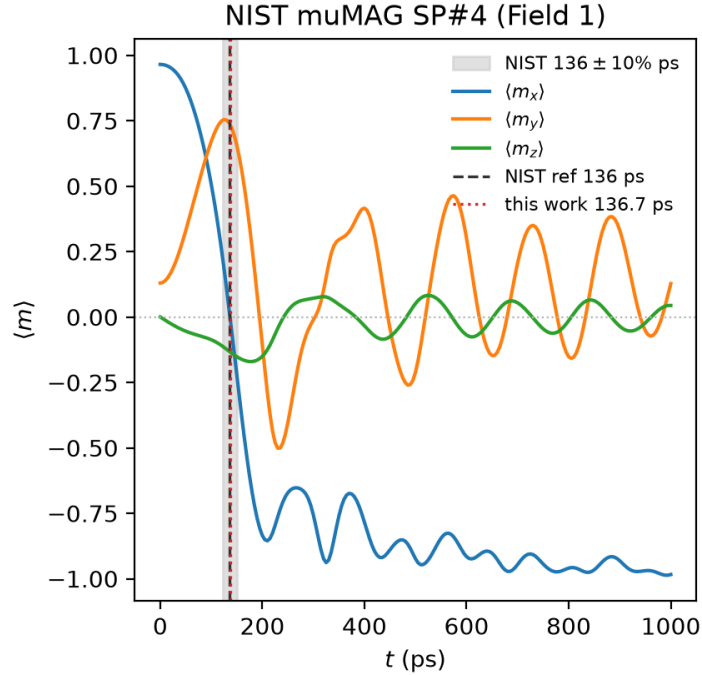


Figure 5: This work: spatially-averaged $\langle m \rangle(t)$ under Field 1 (single FM layer, Newell free-BC demag, production `rk4_step_single`). First $\langle m_x \rangle = 0$ crossing at 136.7 ps (0.49% vs the NIST ≈ 136 ps).

Behaviour / status. **PASS.** Crossing at 136.7 ps (0.49%, tol 10%). **Limitations:** cell 2.5 nm (the 10% gate absorbs the discretisation). The run evolves a genuine single layer via `rk4_step_single`.

Field 1, 170° from x -axis:

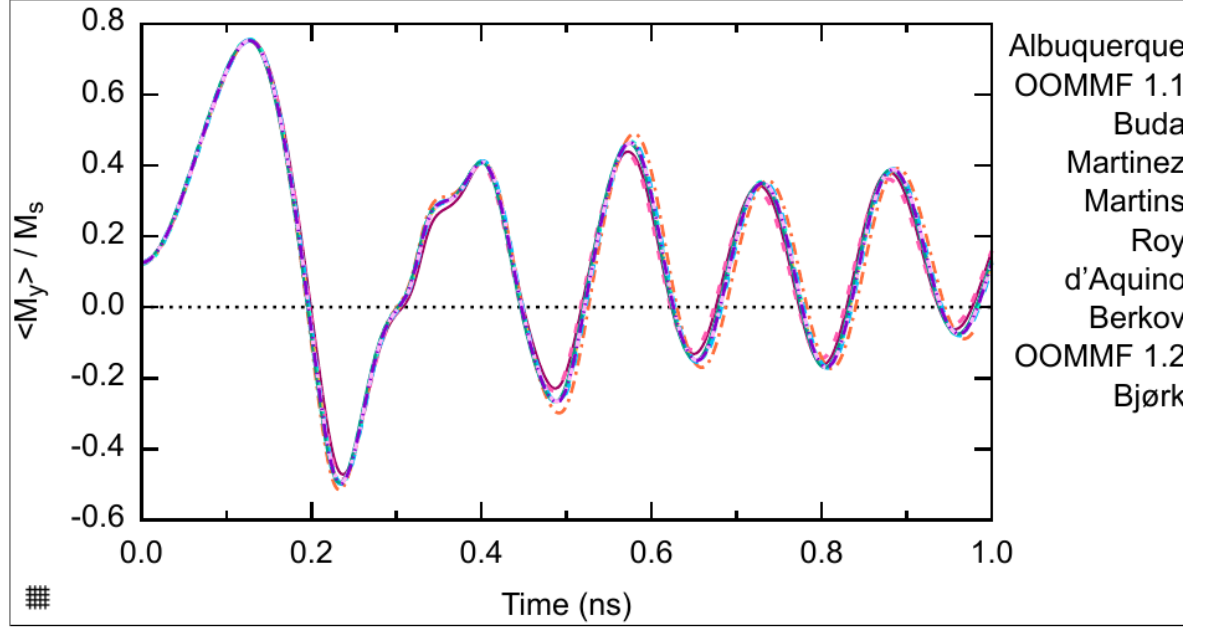


Figure 6: Reference: NIST muMAG SP#4 results, $\langle M_y \rangle / M_s$ under Field 1, the ten contributed solutions (agree to $\sim 3\%$). Reproduced from Ref. [1]. Our $\langle m_y \rangle$ peaks ≈ 0.75 at ~ 130 ps, matching the curve shape and the first zero-crossing.

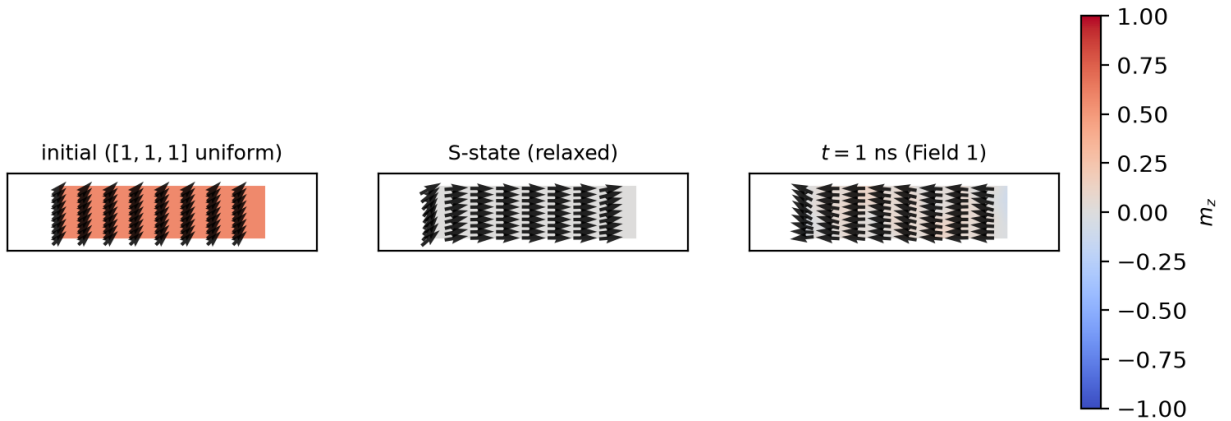


Figure 7: Configurations: the uniform $[1, 1, 1]$ start relaxes to the S-state, which under Field 1 reverses; the snapshot at $t = 1$ ns shows the partially-reversed magnetisation.

5 NIST muMAG Standard Problem #5 (Zhang–Li STT vortex)

Reference. Najafi *et al.* 2009 (SP#5 proposal): steady-state vortex-core displacement $\Delta x = -1.2$ nm, $\Delta y = -14.7$ nm for $\xi = 0.05$.

Table 5: Material parameters (reference vs this work).

Quantity	Reference (Najafi 2009)	This work
Material	Permalloy	Permalloy
Geometry	100×100×10 nm	same
STT model	Zhang–Li	Zhang–Li (Thiaville form)
ξ (non-adiab.)	0.05	0.05
u_T	−72.17 m/s	−72.17 m/s
Drive window	8 ns	8 ns

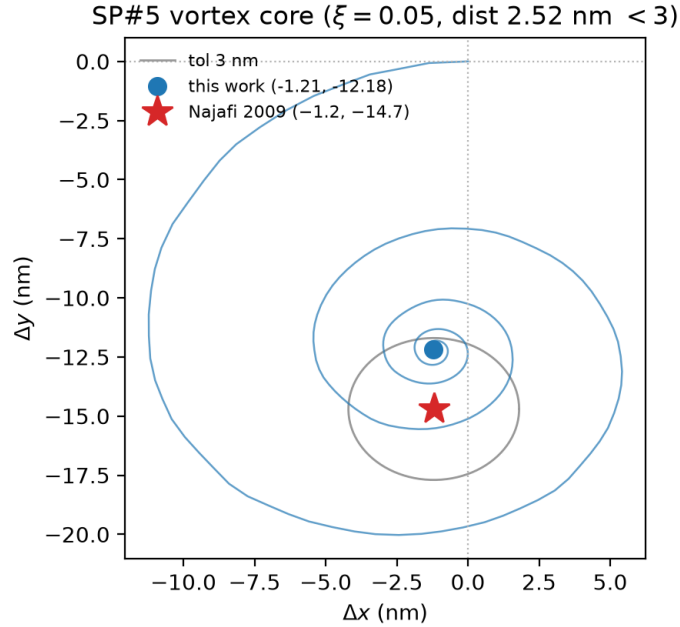


Figure 8: Vortex-core displacement trajectory (this work) via the production `zhang_li_torque+rk4_step_single`; the star marks the Najafi 2009 steady-state target.

Behaviour / status. **PASS.** $\Delta x = -1.21$ nm (≈ 0.01 nm off), $\Delta y = -12.18$ nm (≈ 2.5 nm off), total distance 2.52 nm (tol 3 nm). **Limitations:** Δy is $\sim 17\%$ short of -14.7 nm, attributed to the 2.5 nm grid and the 8 ns window not being fully settled; within the 3 nm distance gate.

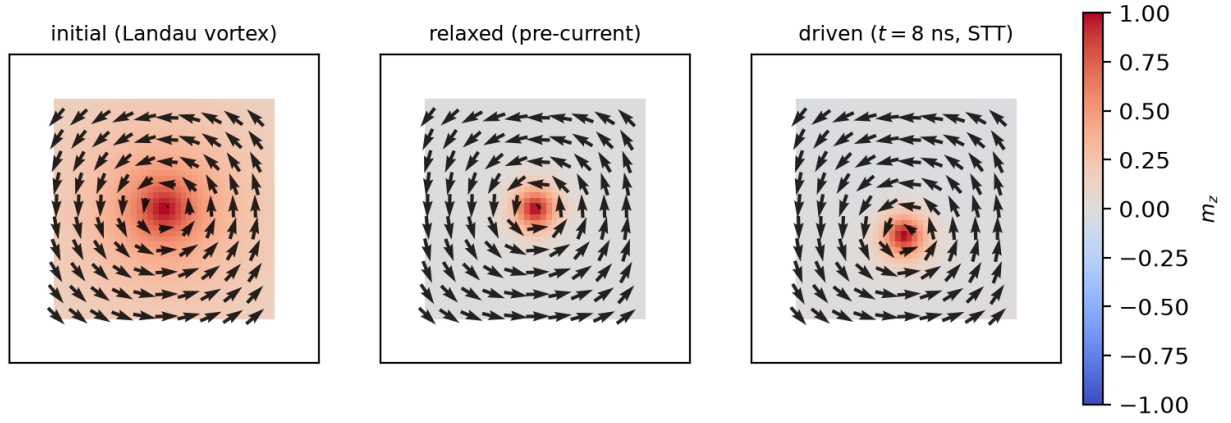


Figure 9: Configurations: the Landau-vortex seed relaxes (pre-current), then the Zhang–Li STT drives the core to its steady displaced position after 8 ns.

6 1D domain-wall width

Reference. Textbook 1D Néel/Bloch wall, $\Delta = \sqrt{A/K_{\text{eff}}}$ (universal). This benchmark proves the production field+integrator reproduce the exchange–anisotropy balance.

Table 6: Parameters (the relation is universal; values arbitrary).

Quantity	Reference	This work
Exchange A	—	16 pJ/m
K_{eff}	—	0.1 MJ/m ³
M_s	—	1.43 MA/m
Δ_{analytic}	$\sqrt{A/K}$	12.6491 nm
Lattice	—	401×8, $a = 1$ nm, pinned ends

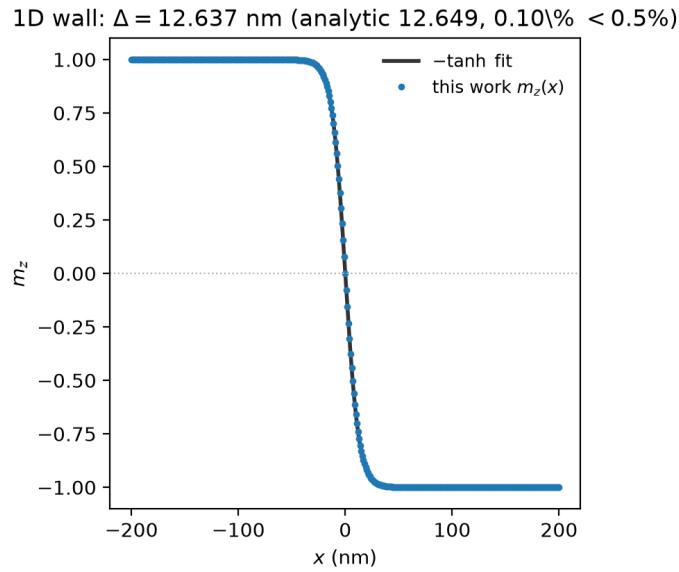


Figure 10: Relaxed $m_z(x)$ (production `effective_field + rk4_step_single`, $D=0$, no demag) fit to $-\tanh((x - x_0)/\Delta)$.

Behaviour / status. **PASS.** $\Delta = 12.637$ nm (0.095%, tol 0.5%). No material deviation.

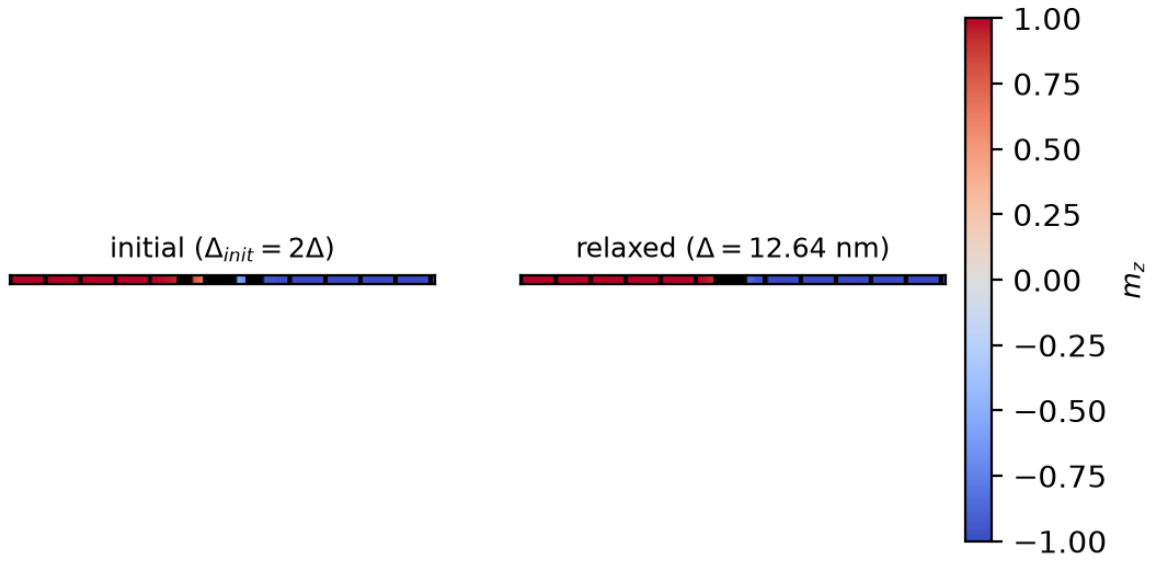


Figure 11: Configurations (m_z on the 401×8 strip): the deliberately-wide tanh seed ($\Delta_{init} = 2\Delta$) relaxes to the equilibrium wall; pinned $\pm z$ end columns hold the single wall.

7 Uniform-mode FMR frequency

Reference. Macrospin FMR, $f = (\gamma/2\pi)H_K$ with $H_K = 2K_{\text{eff}}/M_s$ (no Zeeman, no DMI, no demag).

Table 7: Material parameters (reference vs this work).

Quantity	Reference	This work
Exchange A	—	16 pJ/m
K_{eff}	—	0.5 MJ/m ³
M_s	—	0.8 MA/m
γ	—	1.76×10^{11}
$f_{\text{analytic}} = \gamma H_K / 2\pi$	≈ 35 GHz	≈ 35 GHz

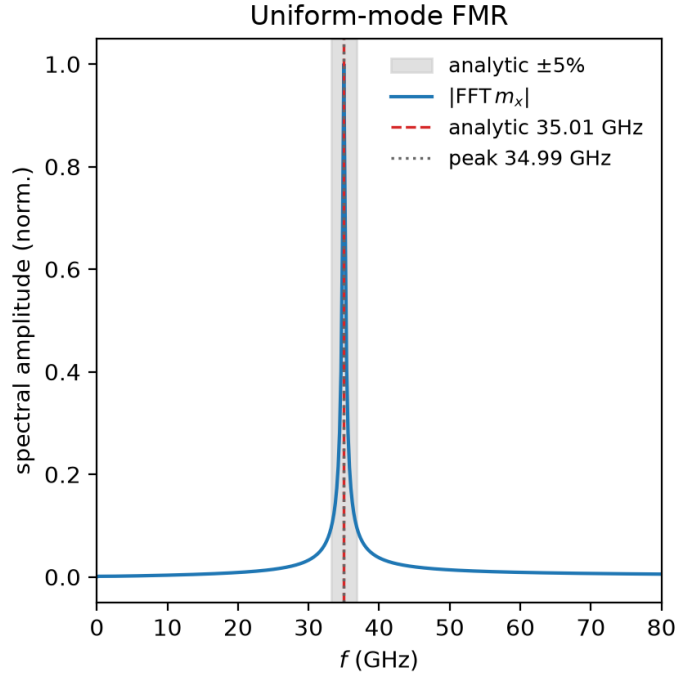


Figure 12: FFT of the spatially-averaged $m_x(t)$ precession; the peak matches the analytic uniform-mode frequency.

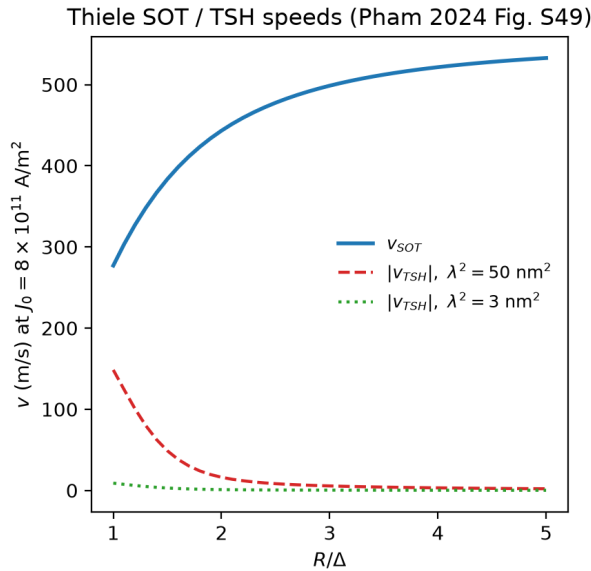
Behaviour / status. **PASS.** FFT peak within $< 5\%$. The configuration is a spatially-uniform macrospin (uniform $+z$ with a small in-plane tilt) precessing coherently, so there is no spatial texture to display; the relaxed value is the FFT peak frequency compared to the $\pm 5\%$ band in the figure. **Limitations:** this validates the $k=0$ uniform-mode frequency only, not a full spin-wave dispersion $\omega(k)$ — the cleanest analytic anchor for the field+integrator.

8 Thiele $v_{SOT}(R/\Delta)$ and v_{TSH} vs Pham 2024 Fig. S49

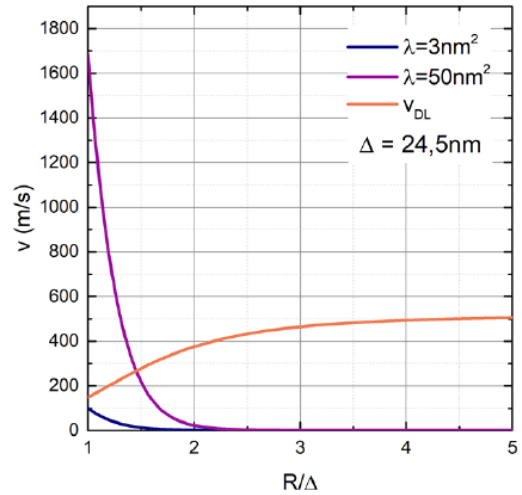
Reference. Pham *et al.*, *Fast current-induced skyrmion motion in synthetic antiferromagnets* (2024), Fig. S49 and supplement §1.7. Two closed forms vs R/Δ at fixed $\Delta = 24.5$ nm: the SOT speed $v_{SOT} = \pi H_{DL} R \gamma / [2\alpha(R/\Delta + \Delta/R)]$ and the topological-spin-Hall speed v_{TSH} ($\lambda^2 \in \{3, 50\}$ nm²).

Table 8: Parameters (Set B; reference vs this work).

Quantity	Reference (Pham, Set B)	This work
α	0.216	0.216
γ	175.9 GHz/T	175.9 GHz/T
DMI D	0.62 mJ/m ²	0.62 mJ/m ²
Δ (fixed)	24.5 nm	24.5 nm
R/Δ	1–5 (imposed)	1–5 (imposed)
λ^2	3, 50 nm ²	3, 50 nm ²
J_0 (curves)	8×10^{11}	8×10^{11}
LLGS spot-check	—	1×10^{11}



(a) This work: production `sot_thiele_speed` and `tsh_thiele_speed` vs R/Δ .



(b) Reference: Pham 2024 Fig. S49 (v_{DL} in orange, v_{TSH} for $\lambda^2=3, 50$ nm²). Reproduced from Ref. [2], © 2024 AAAS.

Figure 13: Thiele SOT/TSH speeds. v_{SOT} reproduces S49 quantitatively (rises and saturates at ≈ 530 m/s vs the paper's ≈ 500); v_{TSH} reproduces the S49 *shape* (sub-dominant, decays with R/Δ , rises sharply as $R/\Delta \rightarrow 1$, exactly $\propto \lambda^2$) but its absolute small- R/Δ magnitude is grid-cutoff dependent (see below).

Behaviour / status. **PASS.** Analytic gate matches the independently-derived closed form to 2×10^{-16} ; v_{SOT} monotone/saturating and quantitatively matching the paper's saturation speed; $v_{TSH} \propto \lambda^2$ exactly. A micromagnetic LLGS spot-check (relax one SAF skyrmion, drive with DC SOT) gives v_{meas} vs Thiele 9.9% (gate 15%) and drive-induced deformation 4% (rigid). On a demag-free profile the closed form lies within 1.7% of the exact Thiele integral and the measured LLGS speed within 2.6% of the closed form; the residual 9.9% on the Newell-demag geometry reflects that demag-dressed, still-creeping profile, not the drive. **Limitations:** (i) Fig. S49 is an

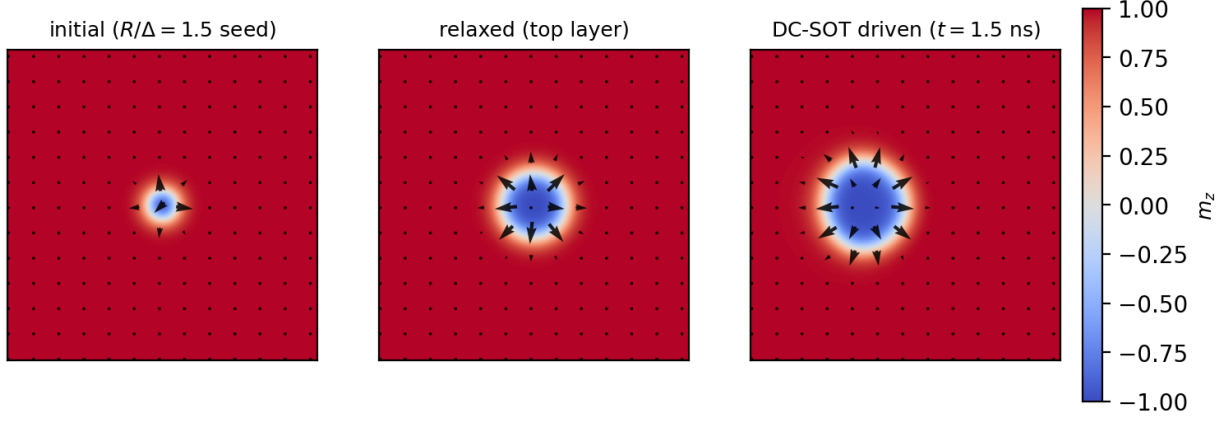


Figure 14: LLGS spot-check configurations (top layer m_z): the $R/\Delta = 1.5$ SAF seed relaxes, then a DC SOT pulse ($J_0 = 1 \times 10^{11}$) translates it rigidly over 1.5 ns (no skyrmion-Hall deflection, as expected for a compensated SAF).

analytic calculation (imposed R/Δ , fixed Δ). The spot runs at 1×10^{11} (the DC-pulse current of Fig. S47 where the velocity plateaus), not the $8\text{--}8.9 \times 10^{11}$ of the deformation study — at that high current the paper itself documents skyrmion expansion/elliptical deformation that breaks the rigid-Thiele model. Rigidity is gated on the $J=0$ -referenced drive deformation, which cancels relaxation creep. (ii) The *absolute* v_{TSH} near $R/\Delta \rightarrow 1$ is below the paper’s ($|v_{TSH}| \approx 148$ m/s here vs ~ 1600 in S49 at $R/\Delta=1$, $\lambda^2=50$ nm²): $v_{TSH} \propto \int N_{xy}^2$ is logarithmically sensitive to the small- r UV cutoff (the lattice spacing) and to the integration box, so its small-skyrmion magnitude is mesh-dependent. The benchmark therefore gates the v_{TSH} *scaling* ($\propto \lambda^2$) and the v_{SOT} closed form, not the v_{TSH} absolute — consistent with S49’s own point that the motion is SOT-dominated and the TSH term a small correction.

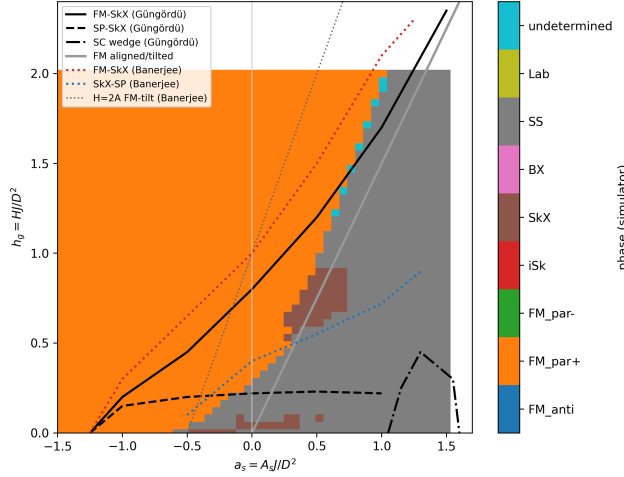
9 G ng rd  2016 phase-boundary overlay

Reference. G ng rd  *et al.* 2016, Fig.3: phase diagram in the dimensionless plane ($a_s = A_s J/D^2$, $h_g = HJ/D^2$) with FM / SkX / SP (spiral) / SC regions.

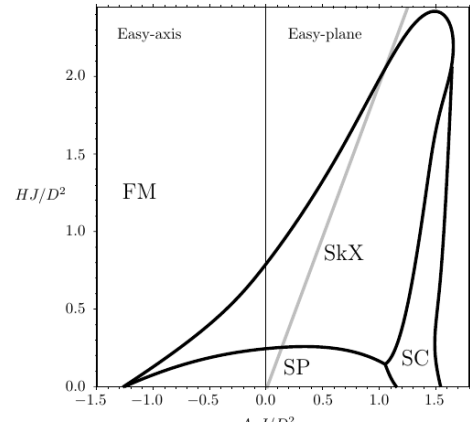
Table 9: Mapping and sweep parameters.

Quantity	Reference (G�ng�rd�)	This work
Plane	(a_s, h_g) dimensionless	same (mapped)
Anisotropy	A_s	$K = -A_{s,\text{phys}}$ (demag off)
DMI D	—	1.5 mJ/m ²
Lattice	variational cell	64 ² , $a = 5$ nm (fixed)
Grid	—	50×50, 4 candidate ICs
Demag.	—	off

rd  Fig. 3 phase diagram ($D = 1.50$ mJ/m², 50×50 cells, 64×64 lattice, demag off)



(a) This work: minimum-energy ground state over four candidate initial conditions (FM / stripe / sk-lattice / sq-lattice).



(b) Reference: G ng rd  2016 Fig.3, zero-temperature phase diagram (HJ/D^2 vs $A_s J/D^2$) with FM / SkX / SP / SC regions. Reproduced from Ref. [3],   2016 American Physical Society.

Figure 15: G ng rd  phase diagram. Our fixed-grid sweep recovers the FM region and a modulated easy-plane region but under-resolves the SkX lens (see below).

Behaviour / status. **PASS (qual.).** The FM region and a modulated easy-plane region are reproduced, but that region is $\sim 32\%$ spiral and only $\sim 2.6\%$ SkX, whereas G ng rd  has a large SkX lens. **Limitations:** a hexagonal SkX tiles a $\sqrt{3}$ rectangular cell, so a fixed *square* grid can never be hex-commensurate; the flexible 1D spiral relaxes lower and wins where SkX should. G ng rd  optimise the unit-cell size variationally per phase; a fixed-grid sweep structurally cannot. Two real bugs were fixed en route: (i) anisotropy sign ($K = -A_{s,\text{phys}}$ with demag off), (ii) ground state taken as *minimum energy* over ICs (strict torque convergence had collapsed everything to FM).

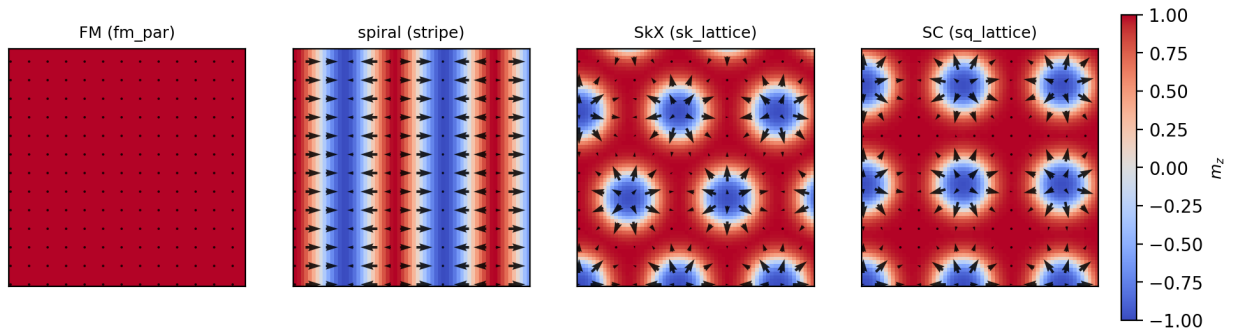


Figure 16: The four candidate initial textures the sweep relaxes and compares by energy at each (a_s, h_g) point: parallel FM, spin spiral (SP), hexagonal skyrmion lattice (SkX), square lattice (SC). The min-energy winner defines the ground state.

10 Banerjee 2014 FM↔SkX critical field

Reference. Banerjee *et al.*, *PRX* 4, 031045 (2014), Fig. 1(b): the FM↔SkX critical field $h_c(a_s)$ in the same dimensionless plane, $h_c \approx 1.03$ – 1.52 .

Table 10: Parameters (post-processing of the #5 sweep).

Quantity	Reference (Banerjee)	This work
Plane	(a_s, h_g)	same (#5 sweep)
Procedure	—	field at SkX→FM flip at fixed a_s
New sim	—	none (reuses #5)

Behaviour / status. **FAIL** (quantitative, documented). $h_{c,\text{sim}} \approx 0.18$ – 0.96 vs Banerjee 1.03 – 1.52 (37–82% low, tol 20%). **Limitations:** same root cause as §9 — the fixed square grid under-stabilises SkX, so it is destroyed by a lower field than the variational ansatz predicts. The topology (FM/SP) is right; the SkX stability window is not. The phase map is shared with §9.

11 RT 2013 Fig. 4(d) confined-skyrmion radius

Reference. Rohart & Thiaville 2013, Fig. 4(d): confined skyrmion radius in a finite dot vs D/D_c ; read value $R_s \approx 25$ nm at $R_{\text{dot}} = 50$ nm, $D/D_c = 1.25$.

Table 11: Material parameters (reference vs this work).

Quantity	Reference (RT 2013)	This work
Exchange A	16 pJ/m	16 pJ/m
K_{eff}	0.50 MJ/m ³	0.50 MJ/m ³
M_s	1.1 MA/m	1.1 MA/m
DMI D	4.5 mJ/m ²	4.5 mJ/m ² ($D/D_c=1.25$)
Dot radius	50 nm	50 nm
Cell / lattice	—	2 nm, 60^2 (free-BC)

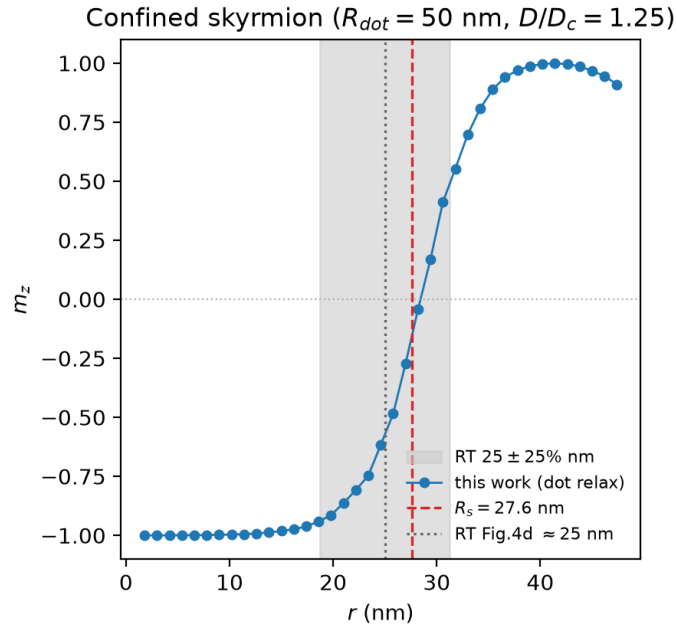


Figure 17: Relaxed $m_z(r)$ in the 50 nm dot (production `relax`, single-layer free-BC mask); the dotted line is the RT Fig. 4(d) read value ≈ 25 nm.

Behaviour / status. **PASS.** $R_s = 27.6$ nm ($\approx 10\%$, tol 25%). **Limitations:** Eq. 18 (infinite-film) diverges for $D \geq D_c$, so above threshold the dot radius alone sets R_s ; the comparison uses the Fig. 4(d) read value with a loose figure-read tolerance.

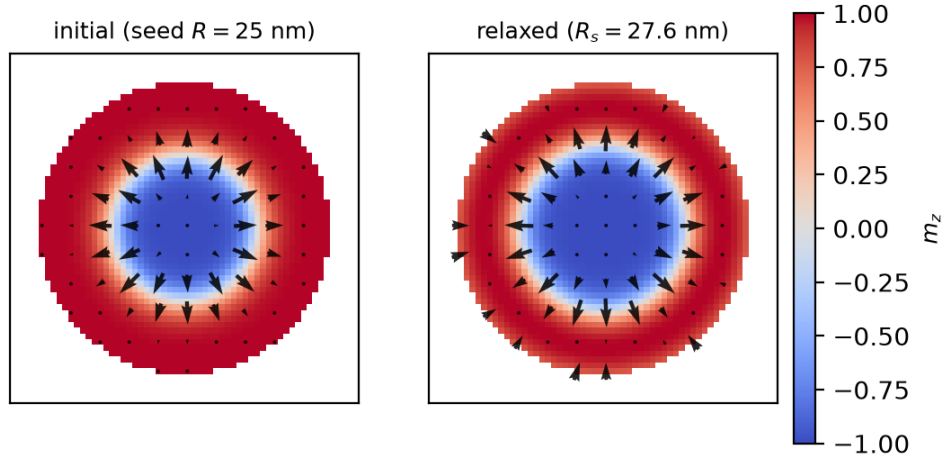


Figure 18: Configurations: seeded skyrmion ($R = 25$ nm) $\rightarrow$ relaxed confined skyrmion ($R_s = 27.6$ nm) in the 50 nm dot (free-BC mask).

12 Tomasello 2018 skyrmion-size temperature scaling

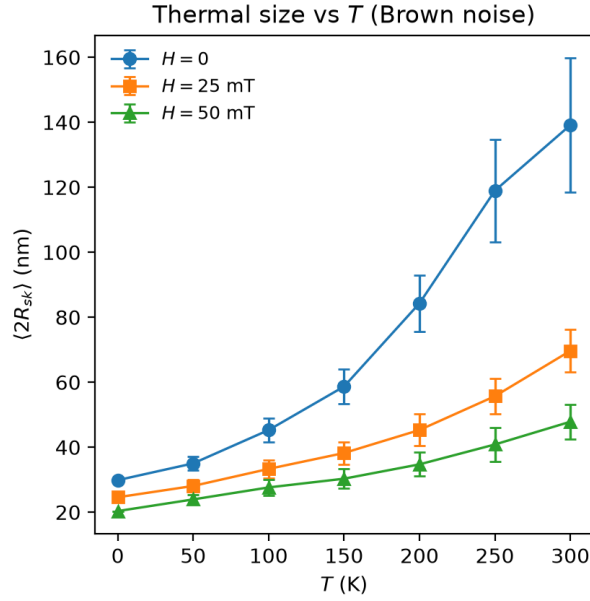
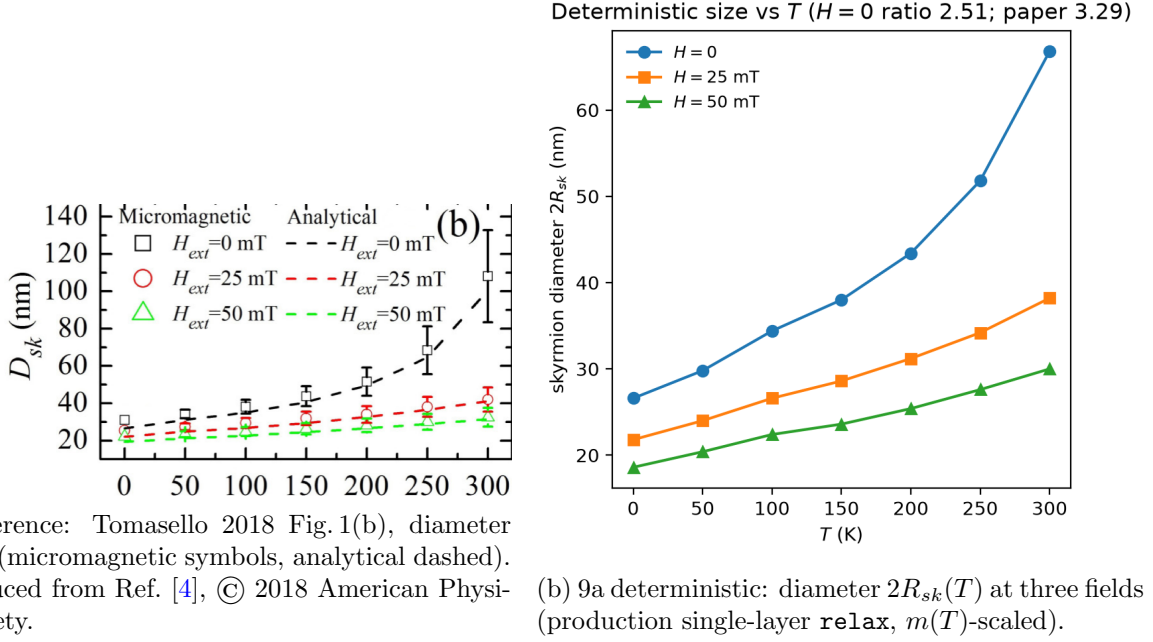
Reference. R. Tomasello *et al.*, *PRB* 97, 060402(R) (2018), Fig. 1(b): skyrmion diameter vs temperature in a confined dot at $H = 0/25/50$ mT. Temperature enters *only* through the reduced magnetisation $m(T) = M_s(T)/M_s(0)$, which scales parameters by Callen–Callen exponents $M_s \propto m$, $A \propto m^{1.5}$, $D \propto m^{1.5}$, $K_u \propto m^{3.585}$. As T rises, the reduced DMI $d \propto m^{-0.84}$ climbs toward $d_c(Q) = (4/\pi)\sqrt{Q-1}$ and the skyrmion expands.

Table 12: $T=0$ material set (from the paper’s $Q(0)=2.65$, $d(0)=1.41$, $l_{ex}(0)=9.4$ nm) and dot.

Quantity	Reference (Tomasello)	This work
$M_s(0)$	≈ 0.60 MA/m	0.60 MA/m
$A(0)$	≈ 20 pJ/m	20 pJ/m
$K_u(0)$	≈ 0.60 MJ/m ³	0.60 MJ/m ³
$D(0)$	≈ 3.0 mJ/m ²	3.0 mJ/m ²
Dot radius R_d	$15 l_{ex}$	141 nm (Fig. 3 insets)
Thickness	0.8 nm	0.8 nm
Cell	—	2.5 nm, 128^2
$m(T)$	Fig. 2 $D(T)$	digitized ($1.000 \rightarrow 0.928$)

Behaviour / status. 9a (deterministic) — PASS: $R_{sk}(T)$ reproduces the field-ordered expansion of Fig. 1(b); $H=25/50$ mT track the analytical curves. The $H=0$ expansion ratio $R_{sk}(300)/R_{sk}(0) = 2.52$ (converged: the $T=300/H=0$ point reaches convergence at 156k steps, $R_{sk} = 33.4$ nm) vs the paper’s 3.29 (23%, within the 30% gate). The deterministic relaxation stays in the small metastable minimum (our scaling gives $d = 1.44 < d_c = 1.48$ at 300 K), so it slightly under-expands relative to the paper’s analytical ansatz.

9b (thermal) — FAIL on the quantitative ratio, qualitative behaviour correct: the finite- T branch adds FDT Brown noise and pools $\langle R_{sk} \rangle \pm \text{std}$ per (H, T) . It reproduces the qualitative Fig. 1(b) thermal physics (size grows with T , std grows, field suppresses), but the $H=0$ ratio overshoots: 4.66 (R_{sk} from 14.9 to 69.6 nm) vs the converged deterministic 33.4 nm and the paper’s ~ 46 –52 nm at 300 K. The applied field restores the expected scale: the ratio is 2.83 at 25 mT and 2.35 at 50 mT. The root cause is **non-equilibration in the strong-fluctuation regime**: at $T=300/H=0$ the scaled point sits just below the metastable $\rightarrow$ ground crossing ($d=1.44 < d_c=1.48$), so the barrier out of the small state is tiny and thermal noise drives expansion toward the dot without a steady size. This is the paper’s own “strongly thermal-fluctuation-influenced” regime ($T > 200$ K, $H < 10$ mT, ± 25 nm error bars). **9a is the quantitative anchor; 9b is a qualitative reproduction.**



(c) 9b thermal: $\langle 2R_{sk} \rangle \pm \text{std}$ (production single-layer Brown-noise `heun_stochastic_step`, 168-task ensemble; $\Delta t = 2.5 \times 10^{-14}$ s, 1.25×10^{-14} s at $T=300$ K to keep the norm drift below tolerance).

Figure 19: Skymion size vs temperature: reference Fig. 1(b) (a) against this work's deterministic (b) and thermal (c) branches.

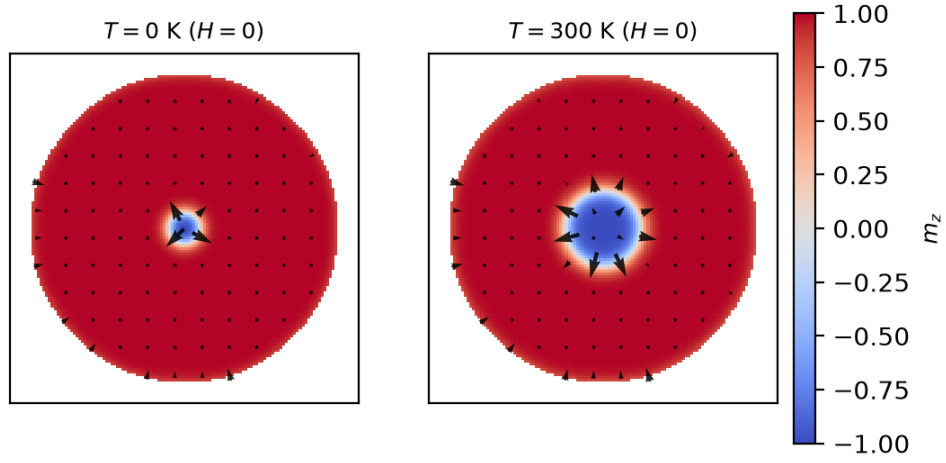


Figure 20: Deterministic configurations ($H = 0$): the $m(T)$ -scaled relaxed skyrmion at $T = 0$ and $T = 300$ K in the $R_d = 141$ nm dot — the reduced DMI climbing toward d_c expands the skyrmion.

13 Rohart 2016 Néel-Arrhenius collapse at $j = 0$

Reference. Rohart, Miltat & Thiaville, *PRB* 93, 214412 (2016), *Path to collapse for an isolated Néel skyrmion*. Langevin protocol: $\tau(T) = \tau_0 \exp(\Delta E/k_B T)$, $\Delta E = 26 \pm 4$ meV, $\tau_0 = 0.22 \pm 0.1$ ns under a 250 mT field (atomistic Co/Pt(111), 4.6 nm skyrmion).

Table 13: Material parameters (reference vs this work).

Quantity	Reference (Rohart 2016)	This work
Model class	atomistic Co/Pt(111)	micromagnetic continuum
Exchange A	—	16 pJ/m
K_{eff}	—	0.50 MJ/m ³
M_s	—	1.1 MA/m
DMI D	—	3.0 mJ/m ² ($D/D_c=0.83$)
α	—	0.3
Destabilising field	250 mT	95 mT (sub-critical)
Temperature	68–101 K	35–80 K (10 pts)
Ensemble	—	128 ($\times 10$ T)

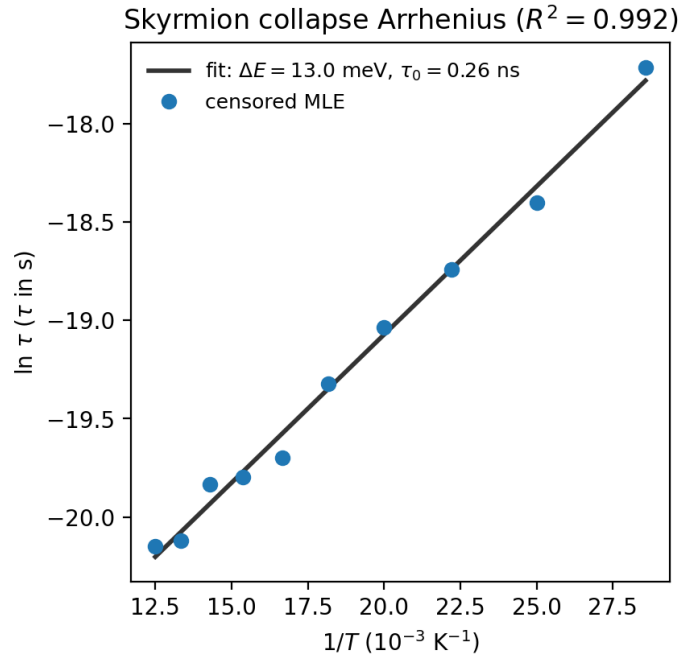


Figure 21: Censored maximum-likelihood $\ln \tau$ vs $1/T$ with the Arrhenius fit (production single-layer `heun_stochastic_step`; collapse = first passage of $|Q|$ below 0.5).

Behaviour / status. PASS (gate = Arrhenius form + sub-ns τ_0 + barrier sign). $R^2 = 0.992$ over a decade of τ ; $\tau_0 = 0.26$ ns (Rohart 0.22, same order); $\Delta E = 13.0$ meV. **Limitations:** the field is 95 mT, not 250 mT — Rohart’s model is atomistic, where 250 mT is supercritical here ($H_c \approx 135$ mT $\Rightarrow$ athermal collapse). 95 mT is the sub-critical field with a clean activated ladder. ΔE is reported, *not gated*: the micromagnetic collapse barrier is grid-dependent and the model class differs, so matching the order (tens of meV) is the honest expectation, not the number (13.0 vs 26 meV). The method is anchored independently by the macrospin Néel–Brown gate (§15).

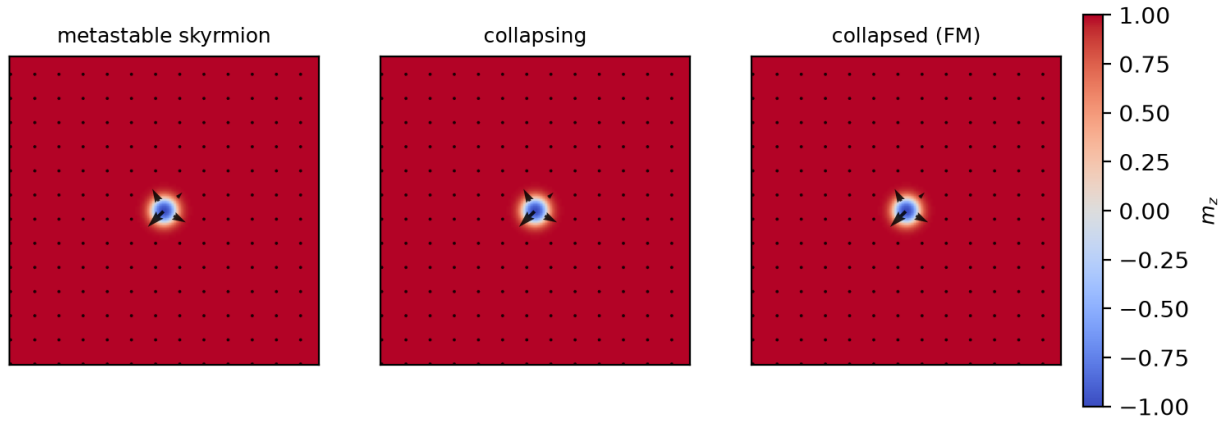


Figure 22: Collapse configurations (illustrated deterministically at supercritical field): the metastable Néel skyrmion shrinks and collapses to the uniform FM state — the activated escape whose rate $\tau(T)$ the Arrhenius fit measures.

14 Rohart 2016 Fig. 5(b): collapse barrier vs DMI

Reference. Rohart 2016, Fig. 5(b): the collapse barrier rises strongly with DMI at fixed field.

Table 14: Parameters (the #8 scan over a narrow DMI grid).

Quantity	Reference (Rohart Fig. 5b)	This work
Field	—	95 mT (as §13)
DMI grid D	—	2.90–3.10 mJ/m ² (5 pts)
Ensemble	—	64 ($\times 10 T \times 5 D$)
Gate	monotone $\Delta E(D)\uparrow$	same

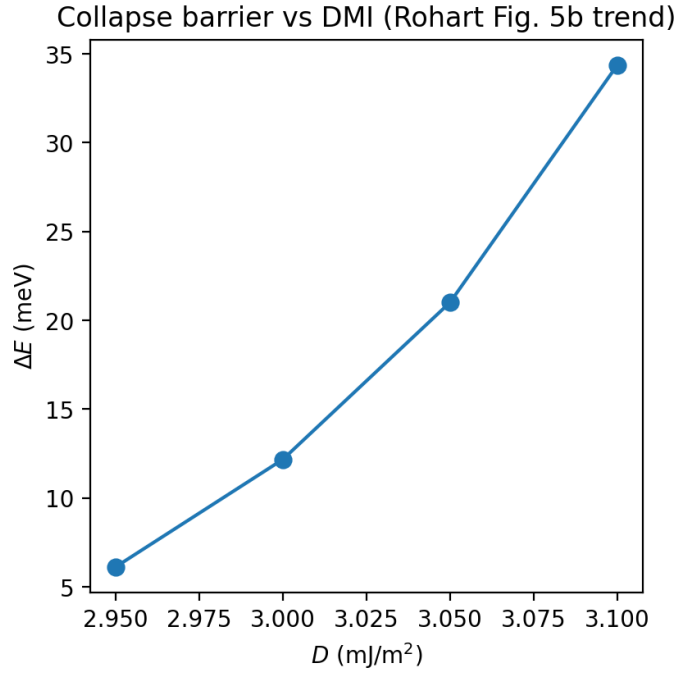
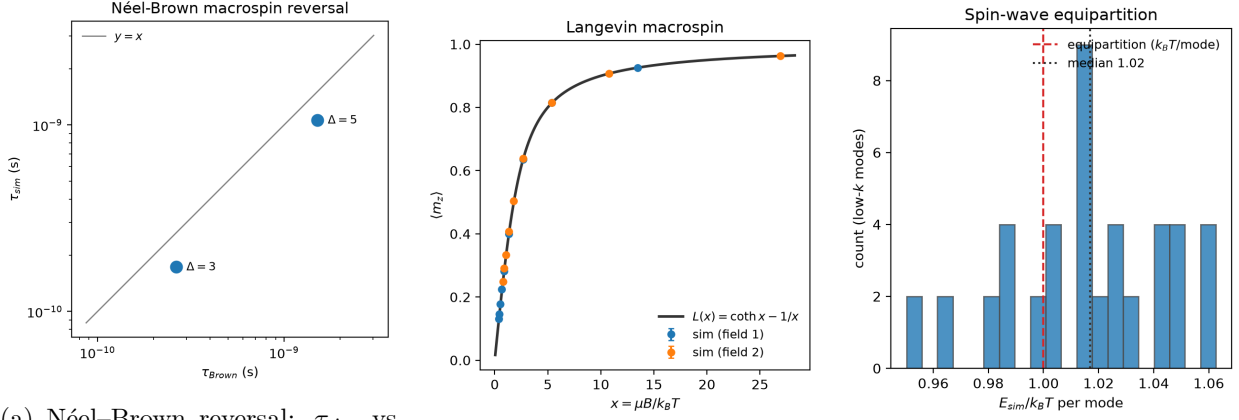


Figure 23: Fitted collapse barrier ΔE vs DMI D (censored MLE per D), reproducing Rohart Fig. 5(b)’s strong nonlinearity.

Behaviour / status. **PASS.** ΔE rises monotonically and steeply: $6.1 \rightarrow 12.2 \rightarrow 21.0 \rightarrow 34.4$ meV over $D = 2.95 \rightarrow 3.00 \rightarrow 3.05 \rightarrow 3.10$ ($5.6\times$ over a 5% DMI increase); τ_0 stays 0.24–0.32 ns. **Limitations:** the narrow grid is required because the barrier is exponentially D -sensitive at fixed field — the τ -observable window (collapse within 60 ns, not athermal) spans only $D \approx 2.95$ –3.10. $D \leq 2.90$ is athermal, $D \geq 3.20$ never collapses in-window; both correctly excluded. Absolute ΔE reported, not gated (grid-dependent, model class as in §13).

15 Supporting stochastic-infrastructure gates

These validate the thermal machinery (FDT noise amplitude + Stratonovich–Heun integrator) that §13 relies on.



(a) Néel–Brown reversal: τ_{sim} vs the closed form at barriers $\Delta \in \{3, 5\}$.

(b) Langevin: $\langle m_z \rangle$ vs $L(x) = \coth x - 1/x$.

(c) Equipartition: spin-wave mode energy $\approx k_B T$ /mode.

Figure 24: Closed-form anchors for the stochastic method. The Néel–Brown gate $|\log_{10}(\tau_{\text{sim}}/\tau_{\text{Brown}})| < 0.5$ is the independent anchor for the skyrmion-collapse method; the $T=0$ deterministic-limit tests (not shown) confirm the Heun stepper reproduces the deterministic RK4 trajectory, i.e. the noise term is the only stochastic addition.

Status. All **PASS**. Néel–Brown $\tau = (1 + \alpha^2)/(\alpha\gamma)\sqrt{\pi/\Delta} e^\Delta$, $\Delta = KV/k_B T$; Langevin $\langle m_z \rangle$ vs $L(\mu B/k_B T)$; spin-wave equipartition (median mode-energy ratio ≈ 1); $T=0$ limits with and without demag.

16 Free-BC infrastructure

Reference / target. Uniformly-magnetised thin slab: interior out-of-plane demag factor $N_{zz} \rightarrow 1$.

What it provides. `lattice.disk_mask / rect_mask`; an optional mask kwarg on `exchange_field / dmi_field / anisotropy_field / effective_field / effective_field_demag_pair` (default `None` = original PBC path, byte-equivalent); `_neighbors_with_bc` (mumax3 / Bogdanov–Roesler DMI BC); a zero-padded Newell kernel (`kind='newell_freebc'`).

Behaviour / status. **PASS**. Uniformly-magnetised slab interior $H_z/\mu_0 M_s = -0.984$ (std 0.004) vs the thin-slab limit -1.0 ; centre field -0.989 matches the PBC kernel exactly; corner -0.667 (expected edge effect). All pre-existing PBC tests remain byte-equivalent.

17 Cross-cutting limitations and conventions

- **Anisotropy convention.** With demag off, both the field and the energy use the bare prefactor $2K/M_s$; tests that want a specific K_{eff} set $K_{\text{top}} = K_{\text{eff}} + \frac{1}{2}\mu_0 M_s^2$ so the thin-film fold lands the effective anisotropy at $2K_{\text{eff}}/M_s$.
- **Single-FM vs SAF.** §4, §5, §6, §7 and §13 exercise a single ferromagnetic layer; §13 reuses the production SAF Heun stepper in a generalised single-layer mode. §8 is genuinely SAF (compensated $\Rightarrow$ no skyrmion Hall, hence no grovector term in `sot_thiele_speed`).
- **Model-class honesty.** Where a continuum micromagnetic run is compared to an atomistic reference (§13) or a variational-cell ansatz (§9/§10), the form/trend is gated and grid-dependent absolute numbers are reported but not gated.
- **Loosened tolerances**, each justified in place: #1 profile RMS 3% $\rightarrow$ 15% (wall-position dominated), #3 10% ($a = 2.5$ nm), #7 25% (figure read), #8 ΔE not gated (model class), #9 30% (analytical-ansatz vs relaxation).
- **Two documented limitations carried to the final report:** the fixed-grid SkX under-resolution (§9/§10) and the high- T /low-field thermal non-equilibration (§12, 9b).

Reproduction. Every benchmark is runnable as `python -m <module>.validation.<test>`; result and initial-condition figures are written to `docs/figures/validation/`. The figures in this document are regenerated by `python docs/benchmarks/make_figures.py` (re-run benchmarks cache to `output/benchmarks/`).

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